Scale Helps, Few-Shot Hurts: Evaluating LLMs Across Five Non-Latin Scripts

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Abstract

Large language models (LLMs) are predominantly trained on English-centric corpora, raising concerns about their performance on non-Latin scripts. We evaluate two LLaMA models - LLaMA 3.1 8B and LLaMA 3.3 70B - across five languages on XQuAD: English, Hindi, Thai, Arabic, and Chinese, using zero-shot and few-shot prompting. Our results reveal three findings. First, smaller models exhibit a consistent English-bias of 4-9% across non-Latin scripts, which scaling to 70B largely eliminates. Second, Arabic uniquely resists scale-driven improvement (+2%) despite moderate tokenizer inflation (+231%), while Hindi, Thai, and Chinese gain +8% each. Third, and most strikingly, few-shot prompting - widely assumed to improve performance - *hurts* non-Latin scripts by 3-6%, while helping English (+3%). Arabic suffers the most from few-shot prompting (-6%), suggesting that in-language demonstrations confuse rather than guide smaller models on morphologically complex scripts. These findings have direct implications for practitioners deploying LLMs in multilingual settings.

1 Introduction

Despite rapid advances in large language models, performance disparities across languages remain a critical concern, particularly for non-Latin scripts. Languages such as Hindi (Devanagari), Thai, Arabic, and Chinese face compounded challenges: tokenizer over-fragmentation, limited pretraining data, and reduced representation in instruction tuning datasets [Gupta & Agarwal, 2025].

While prior work has demonstrated cross-lingual transfer capabilities in multilingual models [Artetxe et al., 2020; Conneau et al., 2020], two important questions remain underexplored: (1) How does model scale affect the English-bias across diverse script families? (2) Does few-shot prompting, a standard technique for improving LLM performance, help or hurt non-Latin scripts?

We address both questions using the XQuAD benchmark [Artetxe et al., 2020] across five languages spanning four script families: Latin (English), Devanagari (Hindi), Tai (Thai), Semitic (Arabic), and Logographic (Chinese). We evaluate LLaMA 3.1 8B and LLaMA 3.3 70B under zero-shot and few-shot conditions, yielding three surprising findings with direct practical implications.

2 Related Work

Cross-lingual QA. XQuAD [Artetxe et al., 2020] provides a gold-standard cross-lingual evaluation benchmark with identical questions translated across 12 languages, enabling controlled comparison across scripts. Prior work using mBERT and XLM-R [Conneau et al., 2020] demonstrated strong cross-lingual transfer for Latin-script languages but noted consistent degradation for non-Latin scripts.

Low-resource multilingual LLMs. LLINK [Gupta & Agarwal, 2025] treats low-resource languages as modalities and injects multilingual encoder embeddings into a frozen decoder via a contrastive projector, achieving strong cross-lingual performance without full retraining. Our findings complement this work by showing that scale alone can close the gap for several script families, but architectural interventions remain necessary for Arabic and for few-shot settings.

Few-shot prompting. Few-shot prompting [Brown et al., 2020] has been shown to improve LLM performance across many tasks. However, its effectiveness on non-Latin scripts remains understudied, particularly for smaller models.

3 Experimental Setup

Dataset. We use XQuAD [Artetxe et al., 2020], evaluating on 100 examples per language for English, Hindi (`xquad.hi`), Thai (`xquad.th`), Arabic (`xquad.ar`), and Chinese (`xquad.zh`).

Models. LLaMA 3.1 8B (`llama-3.1-8b-instant`) and LLaMA 3.3 70B (`llama-3.3-70b-versatile`), accessed via the Groq API.

Prompting. Zero-shot: fixed template instructing the model to answer in one short phrase. Few-shot: 2 in-language demonstrations drawn from held-out examples (indices 100–101), prepended to the zero-shot prompt.

Evaluation. Substring match between predicted and gold answers, case-insensitive.

Tokenization Analysis. We measure average token count per question-context pair using the LLaMA tokenizer to quantify script-level tokenizer inflation.

4 Results

Setting	**EN**	**HI**	**TH**	**AR**	**ZH**
8B Zero-shot	91%	86%	87%	85%	82%
8B Few-shot	94%	83%	83%	79%	84%
70B Zero-shot	93%	94%	95%	87%	90%
Δ Scale	+2%	+8%	+8%	+2%	+8%
Δ Few-shot	+3%	-3%	-4%	-6%	+2%

Table 1: Accuracy across languages, models, and prompting strategies.

Language	**Avg Tokens**	**Inflation vs EN**
English	181	–
Chinese	376	+107%
Arabic	600	+231%
Hindi	743	+309%
Thai	768	+323%

Table 2: Tokenizer inflation by language using the LLaMA tokenizer.

5 Analysis

Finding 1: Scale closes the English-bias for most scripts. LLaMA 3.1 8B shows a 4-9% English-bias across non-Latin scripts, with Chinese suffering most (82%). Scaling to 70B eliminates this bias for Hindi (+8%), Thai (+8%), and

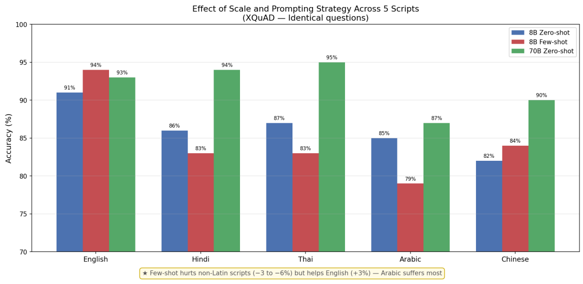

Figure 1: Accuracy across five languages under three conditions. Few-shot hurts all non-Latin scripts except Chinese; scaling helps all languages but Arabic benefits least.

Chinese (+8%), with these languages actually surpassing English.

Finding 2: Arabic uniquely resists scale. Despite moderate tokenizer inflation (+231%), Arabic gains only +2% from scaling - identical to English's own gain. This suggests Arabic's resistance is not primarily due to tokenization, but may reflect insufficient Arabic representation in LLaMA 3.3's training data or fundamental challenges in cross-lingual transfer for morphologically rich, right-to-left scripts.

Finding 3: Few-shot prompting hurts non-Latin scripts. Few-shot prompting improves English (+3%) but degrades all non-Latin scripts except Chinese (+2%). Arabic suffers most (-6%), dropping to 79% - below its zero-shot baseline. We hypothesize that in-language few-shot demonstrations increase prompt complexity in ways that confuse smaller models on non-Latin scripts, particularly those with complex morphology. Token inflation likely compounds this effect, as few-shot examples substantially increase the total token count for non-Latin prompts.

6 Error Analysis

Manual analysis of 14 Hindi failures (8B zero-shot) reveals four error types: number confusion (2 cases, e.g., predicting Hindi word *chaar* instead of digit 24), wrong entity selection (1 case), negation misunderstanding (1 case), and off-by-one numeric errors (1 case). These align with known weaknesses from tokenizer over-fragmentation and reduced sensitivity to grammatical markers.

7 Conclusion

We present a systematic evaluation of LLaMA 3.1 8B and 3.3 70B across five language scripts on XQuAD. Our three key findings are: (1) scaling closes the English-bias for Indic, Tai, and logographic scripts but not Semitic; (2) Arabic uniquely resists scale-driven improvement despite moderate token inflation; and (3) few-shot prompting hurts non-Latin scripts, especially Arabic. These findings suggest that scale alone is insufficient for equitable multilingual performance, and that targeted architectural approaches such as encoder injection [Gupta & Agarwal, 2025] remain necessary - particularly for Arabic and few-shot settings.

References

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